

Assignment: Logic and Proofs

Instructions. All answers must be **handwritten** on separate paper. Write legibly and label each answer with its problem and part number. Use the definitions and methods from the lectures. Show your reasoning whenever requested; a numerical example alone does not prove a universal claim. AI tools are permitted under the course policy, but you are responsible for understanding and checking everything you submit.

1. Key concepts from the course

Which concept do you consider the most important from the material covered so far in Part 1 (Logic) and Part 2 (Proofs)?

2. Propositions and open sentences

Classify each item as a **proposition**, an **open sentence**, or **neither**. For a proposition, also give its truth value. Give a brief reason for each answer. All variables are unassigned unless explicitly fixed or quantified. An open sentence here has a precise truth condition once its free variables are assigned.

- (a) $7 + 5 = 12$.
- (b) 9 is a prime number.
- (c) $x + 3 = 8$, where $x \in \mathbb{Z}$.
- (d) $x^2 \geq 0$, where $x \in \mathbb{R}$ is free.
- (e) For every real number x , $x^2 \geq 0$.
- (f) There exists an integer n such that $n^2 = 2$.
- (g) $x < y$, where $x, y \in \mathbb{Z}$ are free.
- (h) Let $x = 4$. Then $x + 3 = 7$.
- (i) Please sort this list in increasing order.
- (j) Is 17 prime?
- (k) This classifier is good. (No criterion for “good” is specified.)
- (l) A classifier correctly labels 92 of the 100 examples in a fixed test set. Its accuracy on that set is at least 0.90.

3. From a description to a truth table

A data record is *flagged for review* exactly when its label is missing or its value is an outlier, including when both conditions hold.

- (a) Define propositional variables L and O for the two conditions. Express the condition for flagging a record using these variables.
- (b) Complete the following table, using your expression as the heading of the last column.

L	O	Your expression from (a): _____
T	T	
T	F	
F	T	
F	F	

Explain whether a record with both conditions should be flagged, and how the table captures this.

4. Implication and logical equivalence

- (a) State what it means for two propositional forms to be *logically equivalent*.
- (b) Prove $P \Rightarrow Q \equiv (\neg P) \vee Q$ by completing the table below and comparing the relevant columns.

P	Q	$P \Rightarrow Q$	$\neg P$	$(\neg P) \vee Q$
T	T			
T	F			
F	T			
F	F			

- (c) Explain in words why $P \Rightarrow Q$ is true when P is false, for either truth value of Q .
- (d) A policy states: “A dataset enters training only if it passes validation.” Let T mean that the dataset enters training and V mean that it passes validation.
- Express the policy as an implication.
 - Rewrite it using only $\neg$ and $\vee$.
 - Negate the policy and simplify your answer using $\wedge$ and $\neg$. Describe exactly what a policy violation looks like.
 - Does passing validation alone guarantee that the dataset enters training? Justify your answer with a truth assignment consistent with the policy.

5. De Morgan's laws

- (a) Prove each equivalence using a complete truth table:

$$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q), \quad \neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q).$$

- (b) Let A mean “the file is archived” and B mean “the file is backed up.” Write each sentence symbolically and then rewrite it using De Morgan's laws:
- It is not the case that the file is archived or backed up.
 - It is not the case that the file is both archived and backed up.
- (c) Explain why “neither A nor B ” and “not both A and B ” have different meanings. Give a truth assignment on which they differ.

6. Validity is a property of reasoning

- Define a *valid argument form* and an *invalid argument form*. Your definitions must refer to the truth values of *all premises* and the conclusion.
- Complete all four rows. Use $P \wedge Q \therefore P$ for the valid rows and $P \vee Q \therefore P$ for the invalid rows. Specify actual truth values of P, Q producing the requested conclusion.

Argument form	Conclusion	P	Q	Premise truth value
Valid	True			
Valid	False			
Invalid	True			
Invalid	False			

- A student says, “The conclusion is true, so the argument is valid.” Explain the error using one row of your table.
- Another student says, “The conclusion is false, so the argument is invalid.” Explain the error using a different row.
- If an argument is valid and its conclusion is false under a particular truth assignment, what must be true about its premises under that assignment? Explain.

7. Why validity matters

A team uses the rule “If a data file passes validation, it may be used for training.” Let V mean “the file passes validation” and U mean “the file may be used for training.” Treat $V \Rightarrow U$ as a premise.

Two reviewers reason as follows:

Reviewer A: The file passes validation. Therefore, it may be used for training.

Reviewer B: The file may be used for training. Therefore, it passes validation.

- Write each reviewer’s complete argument symbolically.
- Determine which argument is valid. For the invalid one, give a truth assignment making all premises true and the conclusion false.
- Suppose Reviewer B’s conclusion happens to be true for today’s file. Explain why that does not justify reusing B’s reasoning for other files.
- Give an additional implication that would make B’s conclusion follow. Explain why the original rule alone does not supply this implication.
- Explain the two separate things that must be checked before a deductive argument can guarantee a true conclusion.

8. Recognizing rules of inference

For each argument, fill in the conclusion that follows by one rule from the list and name that rule. Use each rule exactly once.

Modus ponens; modus tollens; hypothetical syllogism; disjunctive syllogism; addition; simplification; conjunction.

- (a) $P, P \Rightarrow Q \therefore$ _____
- (b) $P \Rightarrow Q, \neg Q \therefore$ _____
- (c) $P \Rightarrow Q, Q \Rightarrow R \therefore$ _____
- (d) $P \vee Q, \neg P \therefore$ _____
- (e) $P \therefore$ _____ (Use Q in your conclusion.)
- (f) $P \wedge Q \therefore$ _____ (Give one conjunct.)
- (g) $P, Q \therefore$ _____

Then give a short, line-by-line deduction of R from

$$P \vee Q, \quad \neg Q, \quad P \Rightarrow R.$$

State the reason for every line.

9. Valid or invalid?

Classify each argument form as **valid** or **invalid**. For a valid form, name an applicable rule or give a short justification. For an invalid form, specify a truth assignment making *every premise true and the conclusion false*. The symbol $\therefore$ separates the premises from the conclusion.

- (a) $P \Rightarrow Q, P \therefore Q.$
- (b) $P \Rightarrow Q, Q \therefore P.$
- (c) $P \Rightarrow Q, \neg P \therefore \neg Q.$
- (d) $P \Rightarrow Q, \neg Q \therefore \neg P.$
- (e) $P \vee Q, \neg P \therefore Q.$
- (f) $P \vee Q, P \therefore \neg Q.$
- (g) $P \wedge Q \therefore P.$
- (h) $P \therefore P \wedge Q.$
- (i) $P \Rightarrow Q, Q \Rightarrow R \therefore P \Rightarrow R.$
- (j) $P \Rightarrow R, Q \Rightarrow R \therefore (P \vee Q) \Rightarrow R.$

10. Predicates, witnesses, and counterexamples

For parts (a)–(h), let $U = \{-2, -1, 0, 1, 2\}$ and define

$$P(x) : x > 0, \quad E(x) : x \text{ is even}, \quad S(x) : x^2 = 1.$$

Determine whether each proposition is **true** or **false**. For a true existential statement, give a witness; for a false universal statement, give a counterexample. In the other cases, explain why every element satisfies the predicate or why no witness exists, as appropriate.

- (a) $(\exists x \in U)P(x)$.
- (b) $(\forall x \in U)P(x)$.
- (c) $(\exists x \in U)(x = 2)$.
- (d) $(\forall x \in U)E(x)$.
- (e) $(\exists x \in U)S(x)$.
- (f) $(\forall x \in U)(x^2 \geq 0)$.
- (g) $(\exists x \in U)(x^2 = 3)$.
- (h) $(\forall x \in U)(P(x) \Rightarrow x^2 > 0)$.

For parts (i)–(l), the universe is $\mathbb{Z}$. Determine the truth value and justify your answer.

- (i) $(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})(x < y)$.
- (j) $(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})(x < y)$.
- (k) $(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})(x + y = 0)$.
- (l) $(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})(x + y = 0)$.

Explain why an existential statement does not require *exactly one* witness, and why one successful example usually does not establish a universal statement.

11. Translating quantified statements

For (a)–(d), let S be a nonempty set of students and A a nonempty set of assignments. Define a predicate $C(s, a)$ meaning “student s completed assignment a .” Translate each sentence using C , quantifiers, and logical connectives. Make the scope of each quantifier explicit.

- (a) Every student completed at least one assignment.
- (b) There is an assignment that every student completed.
- (c) Some student completed every assignment.
- (d) No student completed every assignment.
- (e) Using $\mathbb{Z}$ as the universe, define suitable predicates and translate: “Every integer whose square is even is even.”
- (f) Negate your formulas in (a) and (e). Move each negation inside all quantifiers and simplify any negated implication. Translate the resulting formulas back into words.

12. Writing proofs and explaining why they work

For **each** part below:

1. State the universe(s), define suitable predicates, and write the full theorem with quantifiers and logical connectives.
2. Give a complete proof using the specified method. State your assumptions and conclusion clearly.
3. Answer the question about why the method is valid.

Use the definitions: an integer is even if it is $2k$ for some $k \in \mathbb{Z}$, and odd if it is $2k + 1$ for some $k \in \mathbb{Z}$. You may use the fact that every integer is either even or odd, but not both.

(a) Direct proof. The sum of any two odd integers is even.

Define predicates O and E for oddness and evenness. Use them to write the quantified theorem. In your proof, use separate integer variables for the two odd integers.

Why does it work? To prove $P \Rightarrow Q$, a direct proof temporarily assumes P and derives Q . Explain why this rules out the only truth assignment that would make the implication false. Also explain what happens when P is false. Do we need to prove that P is actually true?

(b) Proof by contraposition. For every integer n , if n^2 is odd, then n is odd.

Write both the original theorem and its contrapositive with quantifiers. Prove the contrapositive using the definition of an even integer.

Why does it work? State the logical equivalence that justifies contraposition. Explain why proving the *converse* would not serve the same purpose.

(c) Proof by contradiction. There is no smallest positive rational number.

Let $\mathbb{Q}_{>0} = \{r \in \mathbb{Q} : r > 0\}$. Define a suitable two-variable predicate and express the theorem using quantifiers over $\mathbb{Q}_{>0}$. Then write the negation of the entire theorem and use it as the assumption for a contradiction proof. Recall that a rational number has the form a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$.

Why does it work? If assuming $\neg T$ leads to both a statement B and its negation $\neg B$, why does this establish T ? Identify the specific contradiction in your proof.

(d) Proof by cases. For every integer n , $n^2 + 3n$ is even.

Use the two cases “ n is even” and “ n is odd,” and justify the conclusion in each case.

Why does it work? Explain why these cases cover every allowed input. In general, why do $P_1 \vee P_2$, $P_1 \Rightarrow Q$, and $P_2 \Rightarrow Q$ establish Q ? Must the cases in a proof by cases be disjoint?

13. Induction and strong induction

(a) **Mathematical induction.** Prove that for every integer $n \geq 1$,

$$\sum_{i=1}^n (2i - 1) = n^2.$$

Your proof must explicitly state:

1. the proposition $P(n)$;
2. the base case;
3. the induction hypothesis for an arbitrary integer $k \geq 1$;
4. the inductive step, including the exact use of the hypothesis;
5. the final conclusion by induction.

Explain why checking $n = 1, 2, 3$ alone would not prove the theorem.

(b) **Why strong induction follows from ordinary induction.** Let $P(n)$ be a proposition for each integer $n \geq 1$. Suppose that

$$P(1) \text{ is true}$$

and that for every integer $k \geq 1$,

$$(P(1) \wedge P(2) \wedge \cdots \wedge P(k)) \Rightarrow P(k+1).$$

Use *ordinary mathematical induction*, not strong induction, to prove that $P(n)$ holds for every $n \geq 1$.

1. Define $Q(k) = P(1) \wedge \cdots \wedge P(k)$. Explain what $Q(k)$ says.
2. Prove $Q(1)$.
3. Assume $Q(k)$ for an arbitrary $k \geq 1$. Use the given implication to obtain $P(k+1)$, and then prove $Q(k+1)$.
4. Apply ordinary induction to Q and explain why the result implies every $P(n)$.
5. Explain why the strong induction hypothesis is a temporary assumption inside an inductive step, rather than an assumption that the entire theorem is already true.

(c) **Strong induction: forming integer amounts.** Prove that every integer $n \geq 12$ can be written as

$$n = 4a + 5b \quad \text{for some integers } a, b \geq 0.$$

For example, $12 = 4 \cdot 3 + 5 \cdot 0$. Think of a and b as the numbers of 4-unit and 5-unit tokens.

1. Define $P(n)$ using existential quantifiers for a and b .
2. Choose and verify enough consecutive base cases for your proof.
3. State the strong induction hypothesis and complete the proof. You may reduce the target amount by 4; check carefully that the smaller amount lies within the range covered by your hypothesis.
4. Explain why, with this reduction, checking only $n = 12$ would leave a gap.